Stroke Dataset and Python Code Viva

76 questions with short answers, code explanations and a practical demonstration guide

Stroke Risk Factor Analysis Using Patient Health Records

Python Programming Laboratory | EXCP | Academic year 2026-27

Prepared for revision on 8 October 2026

Which version does this guide explain?

The current four-file project: main.py, data_processing.py, statistical_analysis.py and visualization.py, with stroke.csv beside main.py. Answers refer to the actual files in the current Stroke_Analysis_Project folder.

Important difference from your earlier slides and report

This code calculates unadjusted (raw) p-values at alpha = 0.05. It does not calculate Holm correction, Hedges g, Cramer V, Wilson intervals or mean-difference confidence intervals. The earlier report includes those methods. Explain this openly if the examiner compares the code with the report.

Fact to remember	Value
Raw dataset	5,110 records; 12 columns; 249 stroke labels
Primary adults	4,254 records; 247 stroke labels
Observed adult BMI	4,073 records; 208 stroke and 3,865 no-stroke
Known adult smoking	3,392 records; 202 stroke and 3,190 no-stroke
Complete adults	3,256 records; 180 stroke labels
Main scope	Recorded associations; no causal or individual prediction claim

How to practise

Read each question aloud, answer in your own words, then open the named function. Focus first on missing values, Unknown smoking, adult filtering, Welch tests, chi-square tests and raw versus adjusted p-values.

1. Data dictionary and project scope

Source: stroke.csv; data_processing.py constants; Stroke_Capstone_Report.pdf, pages 2-3.

Column	Meaning and role
id	Positive whole-number record identifier; traceability only
gender	Female, Male or Other; categorical
age	Age in years; primary continuous factor
hypertension	0 = absent, 1 = present; primary categorical factor
heart_disease	0 = absent, 1 = present; supplementary correlation variable
ever_married	Yes or No; retained category
work_type	Private, Self-employed, Govt_job, children, Never_worked
Residence_type	Urban or Rural; retained category
avg_glucose_level	Average glucose; recorded unit not independently verified
bmi	Body mass index; conventionally kg/m2; may be missing
smoking_status	formerly smoked, never smoked, smokes, Unknown
stroke	0 = no recorded stroke, 1 = recorded stroke; outcome

1. What is the aim of your project?

Answer: To examine how age, average glucose, BMI, hypertension and smoking status relate to the recorded stroke label. The project cleans the data, compares groups, applies statistical tests and explains limitations.

2. What is CSV?

Answer: CSV means comma-separated values. It stores tabular data as text. Each data row is a record and each column is a variable. Pandas reads stroke.csv into a DataFrame.

3. What are the outcome and explanatory factors?

Answer: stroke is the outcome. The five primary explanatory factors are age, avg_glucose_level, bmi, hypertension and smoking_status. The project does not treat every remaining column as a primary tested factor.

4. Is this a machine-learning prediction project?

Answer: No. The code uses descriptive statistics and hypothesis tests. It does not train a classifier, split data into training and test sets, or calculate prediction accuracy.

2. Understanding stroke.csv

Source: current CSV; make_populations(); population_summary.csv. Values checked from the supplied file.

5. How many records and variables are there?

Answer: There are 5,110 records and 12 columns. The first CSV line contains column names and is not counted as a patient-record row.

6. What do 0 and 1 mean in the stroke column?

Answer: 0 means no recorded stroke and 1 means recorded stroke. The raw dataset contains 4,861 zeros and 249 ones. These are recorded labels, not forecasts.

7. What does the 4.87% stroke percentage mean?

Answer: It is 100 x 249 / 5,110, the fraction of raw records labeled stroke. It is not population incidence or an individual probability of future stroke.

8. Why is age a floating-point variable?

Answer: The dataset includes children with fractional ages. The observed raw age range is 0.08 to 82 years. A fractional age is valid and should not be rounded to an integer during cleaning.

9. Which variables are continuous and which are categorical?

Answer: Age, glucose and BMI are continuous measurements. Hypertension and stroke are binary categories, although stored as numbers. Smoking status, gender, residence and employment categories are categorical strings.

10. Why do you analyze adults separately?

Answer: The main analysis uses age >= 18 because children and adults differ in smoking histories and BMI interpretation. The code retains all ages for descriptions and sensitivity checks instead of deleting child records from the source file.

11. Why do test sample sizes differ?

Answer: Age and glucose use all 4,254 adults. BMI uses 4,073 adults with observed BMI. Smoking inference uses 3,392 adults with known smoking status. Each test excludes only the missing or unknown information relevant to that factor.

12. Can the dataset source and measurements be fully trusted?

Answer: The report identifies the Kaggle benchmark and a GitHub mirror, but hospital, collection dates, sampling and clinical adjudication were not independently verified. The glucose unit and protocol are also unverified. I treat it as an educational benchmark.

3. Missing data, duplicates and outliers

Source: validate_and_clean(); make_populations(); audit_tables(); current CSV.

13. Where are the missing values?

Answer: BMI has 201 missing values across all ages and 181 among adults. Other required columns contain no empty values in the supplied CSV. Unknown smoking is a recorded text category, so isna() does not count it as missing.

14. How does the code recognize N/A BMI cells?

Answer: read_dataset() calls pd.read_csv(file_path, na_values=["N/A"]). Pandas represents those missing measurements as NaN. A missing BMI is not treated as zero.

15. Do you fill missing BMI using the median?

Answer: No. This current implementation performs no BMI imputation. BMI summaries and tests use observed values only. The earlier report describes an optional teaching export, but this four-file code does not create it.

16. Why not change Unknown smoking to never smoked?

Answer: Unknown means the smoking history is not known. never smoked is a specific recorded history. Combining them would introduce an unsupported assumption and alter category counts and test results.

17. What is a complete adult record in this project?

Answer: It means age >= 18, observed BMI and smoking status in the three known categories. There are 3,256 such records. Complete adults form a separate sensitivity population, not the default population for every primary test.

18. How are duplicate records and repeated IDs handled?

Answer: Exact duplicate rows are removed from a working copy if present. After that, any repeated ID indicates different records sharing an identifier, and the code raises ValueError. The supplied CSV has neither exact duplicates nor repeated IDs.

19. What is the IQR outlier rule?

Answer: IQR equals the third quartile minus the first quartile. The code flags values below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR. In adults it flags 0 age values, 594 glucose values and 112 BMI values.

20. Why retain outliers and examine missingness by outcome?

Answer: An unusual measurement may be genuine, so a statistical flag alone is not a reason to delete it. Missing BMI is also unequal: 39/247 stroke records (15.79%) versus 142/4,007 no-stroke records (3.54%). Observed-BMI comparisons may therefore be biased.

4. main.py and the menu

Source: main(), prepare_data(), analyze_factor(), run_complete_analysis() in main.py.

21. Why split the program into four modules?

Answer: data_processing.py reads and validates data. statistical_analysis.py calculates results. visualization.py draws graphs. main.py controls the menu and coordinates the functions. This keeps responsibilities clear and avoids repeating calculations.

22. How does the program locate stroke.csv?

Answer: os.path.abspath(__file__) gives main.py's path. os.path.dirname() gets its folder, and os.path.join() adds stroke.csv. This works when the project moves or the terminal starts from another folder.

23. What happens before the menu appears?

Answer: main() reads the dataset, then prepare_data() validates it and creates the populations. It also checks that the adult and complete-adult populations are not empty. A preparation error stops the program with a readable message.

24. How does the menu keep running?

Answer: while True repeats the menu. input() reads text, int() converts the choice to a whole number, and if/elif selects an operation. Choosing 0 executes break and exits the loop.

25. What is the difference between break, continue and return?

Answer: break leaves the current loop. continue skips to the next loop iteration, such as after invalid menu text. return leaves a function, for example when dataset preparation cannot finish.

26. What happens if the user enters abc or 99?

Answer: int("abc") raises ValueError, which the program catches before returning to the menu. 99 converts successfully but does not match a valid choice, so the program prints an invalid-choice message.

27. What do menu options 8 and 10 do?

Answer: Option 8 calculates, displays and exports the five adult primary tests. Option 10 runs the complete audit, summaries, primary and supplementary tests, sensitivity checks, correlations, report comparison and all charts.

28. Why use if __name__ == "__main__"?

Answer: It calls main() when the file runs directly. If another file imports main.py, the functions remain available without automatically starting the interactive menu. This supports reuse and non-interactive analysis.

5. Explaining the cleaning code

Source: read_dataset(), validate_and_clean(), make_populations(), add_age_bands() in data_processing.py.

```
data = raw.copy()
converted = pd.to_numeric(data[column], errors="coerce")
invalid = data[column].notna() & converted.isna()
adults = data.loc[data["age"] >= 18].copy()
```

29. Why call raw.copy()?

Answer: The analysis works on a separate DataFrame instead of modifying the original in-memory raw data. The code also never writes to stroke.csv. Copies keep population-specific changes separate.

30. Why combine errors="coerce" with an invalid-value check?

Answer: coerce turns unconvertible numeric text into NaN. The following check identifies originally present values that became NaN and raises an error. The code therefore does not silently treat invalid text as a valid missing measurement.

31. What does np.isfinite(observed).all() check?

Answer: It checks that every observed numeric value is finite. Infinite values fail validation. BMI missing values are removed from this finite-value check and handled separately, while missing required non-BMI values are rejected.

32. Which other validation checks are implemented?

Answer: The code checks required columns, binary labels, nonnegative age, positive observed glucose and BMI, positive whole-number IDs, and permitted categorical values. Unexpected values cause a readable error rather than automatic relabeling.

33. Explain data.loc[data["age"] >= 18].copy().

Answer: data["age"] >= 18 creates a True/False mask. .loc selects the rows where the mask is True. .copy() creates an independent adult DataFrame. This is Boolean filtering.

34. How does pd.cut() assign age bands?

Answer: The bins are 18, 30, 40, 50, 60, 70, 80 and infinity. right=False makes intervals left-inclusive and right-exclusive. Age 30 enters 30-39, age 80 enters 80+, and ages below 18 are already excluded from the adult input.

6. Pandas and NumPy in the project

Source: display_dataset_info(), describe_values(), descriptive_table(), proportion_table(), export_tables().

35. What is a DataFrame and what is a Series?

Answer: A DataFrame is a two-dimensional labeled table. A Series is a one-dimensional labeled collection, such as data["age"]. The program reads the CSV as a DataFrame and selects Series for measurements and labels.

36. How do head(), shape, dtypes and isna().sum() help?

Answer: head() displays the first five rows. shape gives row and column counts. dtypes shows how Pandas stores each column. isna().sum() counts missing cells by column. Exact dtype names can vary with the installed Pandas version.

37. What do dropna(subset=["bmi"]) and drop_duplicates() do?

Answer: The first removes rows with missing BMI from the selected working population. The second removes identical full rows. Neither operation changes the source CSV unless the program explicitly writes back, which it does not.

38. Why use .to_numpy(dtype=float)?

Answer: It converts a Pandas Series of observed measurements into a numerical array for NumPy summaries and SciPy tests. dtype=float ensures numerical calculations use floating-point values.

39. How are mean, median and standard deviation calculated?

Answer: describe_values() uses np.mean(), np.median() and np.std(..., ddof=1). The mean is the arithmetic average, the median is the middle value, and standard deviation describes spread around the mean.

40. Why specify ddof=1?

Answer: It calculates sample standard deviation using n - 1 in the variance denominator. NumPy's default ddof=0 uses n. The code needs at least two observations before calculating sample standard deviation.

41. How does groupby().agg(["count", "sum", "mean"]) calculate proportions?

Answer: The code groups records by the chosen category. With a complete binary stroke label, count gives group size, sum gives stroke count and mean gives the stroke fraction. Multiplying the mean by 100 produces the within-category percentage.

42. Why use dictionaries, pd.concat() and index=False?

Answer: Dictionaries label populations and output tables so functions can loop over them. pd.concat() combines test tables into one sensitivity table. to_csv(index=False) avoids exporting an extra numerical index column.

7. Statistical tests and hypotheses

Source: continuous_test(), categorical_test(), conclusion(), primary_tests() in statistical_analysis.py.

43. Which five primary tests are performed?

Answer: Three Welch t-tests compare adult mean age, glucose and BMI. Two Pearson chi-square tests examine hypertension and known smoking categories against stroke status. Heart disease is not a sixth primary test.

44. Why choose Welch rather than the equal-variance t-test?

Answer: The outcome groups have very different sample sizes and may have different variances. stats.ttest_ind(..., equal_var=False, alternative="two-sided") uses Welch's method without requiring equal group variances.

45. What are H0 and H1 for the age test?

Answer: H0 states that the two outcome groups have equal population mean age. H1 states that the means differ. The test is two-sided, so a difference in either direction can count as evidence against H0.

46. What is a p-value?

Answer: Assuming H0 and the test assumptions hold, it is the probability of a test statistic at least as extreme as the observed one. It is not the probability that H0 is true, a patient's stroke probability, or the size of the difference.

47. What does alpha = 0.05 mean here?

Answer: The code uses 0.05 as the threshold for each unadjusted test decision. A raw p-value below 0.05 leads to rejecting H0. Multiple such tests increase false-positive risk because the current code does not apply a family-wise correction.

48. Does p >= 0.05 prove that groups are equal?

Answer: No. It means there is insufficient evidence to reject the specified null at this threshold. The observed adult BMI test does not detect a mean difference, but it cannot establish that BMI has no relationship with stroke.

49. How is the chi-square test built?

Answer: pd.crosstab() creates category-by-outcome counts. The code ensures outcome columns 0 and 1 are present, then calls stats.chi2_contingency(observed, correction=False). H0 is independence and H1 is an association.

50. What assumptions and expected-count checks matter?

Answer: Records should be independent, but this source cannot verify that assumption or random sampling. The code checks chi-square expected counts and warns if the minimum is below 5. Here the minima are 28.86 for hypertension and 46.45 for smoking.

8. Deeper statistics and correlations

Source: categorical_test(), continuous_test(rank_test=True), correlation_tables(), build_tables().

51. How are expected counts and degrees of freedom found?

Answer: Under independence, an expected cell count equals row total x column total / grand total. Chi-square compares observed and expected counts. Degrees of freedom are (rows - 1) x (columns - 1): 1 for hypertension and 2 for known smoking.

52. What does correction=False mean?

Answer: It disables the continuity correction used for a 2 x 2 contingency table. This choice matches the earlier report's Pearson chi-square calculation. It is not a multiple-testing correction and does not adjust the p-values across factors.

53. Why include the Mann-Whitney U test?

Answer: It provides a supplementary rank-based comparison of outcome-group distributions, useful for skewed measurements such as glucose. The code uses alternative="two-sided" and method="asymptotic". It is not automatically a test of medians unless suitable distribution-shape assumptions hold.

54. How do Pearson and Spearman correlations differ?

Answer: Pearson describes linear association between numerical values. Spearman describes association between their ranks and can capture monotonic patterns. The program exports both matrices, while the heatmap displays Spearman correlations.

55. Why exclude IDs and encoded smoking categories from correlations?

Answer: IDs have no clinical measurement meaning. Arbitrary numbers assigned to smoking, gender or work categories would impose an unsupported order. The code includes age, glucose, BMI, hypertension, heart disease and stroke, with binary labels explicitly coded 0/1.

56. What are available pairs, and why export pair counts?

Answer: Each correlation uses records where both selected variables are observed. Age with age uses 4,254 adults, while age with BMI uses 4,073. The pair-count matrix shows these changing denominators. Correlation is not evidence of causation.

9. Graphs and visualization code

Source: visualization.py, including draw_age_histogram(), draw_boxplot(), draw_proportions(), generate_all_charts().

57. What are the roles of Matplotlib and Seaborn?

Answer: Matplotlib provides figures, axes, labels, bars, histograms and saving. Seaborn adds convenient statistical plots and styling, such as boxplots, countplots and heatmaps. Both work with the same prepared data.

58. Which graphs does the program create?

Answer: It creates 10 individual PNGs and one dashboard: outcome counts, age distributions, glucose and BMI boxplots, hypertension and smoking proportions, a Spearman heatmap, age-band proportions, smoking counts, BMI missingness and a 2 x 2 adult dashboard.

59. Why use density=True in the age histogram?

Answer: Each outcome group's histogram area becomes one, allowing comparison of distribution shapes despite the unequal group sizes. Histogram heights represent density, not patient counts or stroke probability.

60. What does a boxplot show?

Answer: It displays the median, quartiles, spread and individually plotted unusual values under the boxplot rule. The program uses observed measurements and keeps outliers visible. A boxplot does not by itself establish statistical significance.

61. Why show percentages and denominators on proportion bars?

Answer: Groups differ in size, so raw stroke counts alone can mislead. A label such as 66/497 = 13.28% gives both the numerator and the relevant category total. Percentage bars start at zero.

62. Why are smoking count and smoking proportion charts different?

Answer: A countplot shows how many adults belong to each smoking category. A proportion chart shows what fraction within each category has the stroke label. Both include Unknown descriptively, although the primary smoking test excludes it.

63. How do tight_layout(), savefig() and close() help?

Answer: tight_layout() adjusts spacing. savefig(..., dpi=180, bbox_inches="tight") saves a clear PNG with a tight bounding box. plt.close(figure) releases the figure after display or saving and reduces accumulated windows and memory use.

64. Why do charts appear one at a time, and do they show confidence intervals?

Answer: save_and_show() calls plt.show() when show=True, so close the current plot to continue. show=False saves without displaying. The current code's proportion bars have no confidence-interval error bars; the earlier report uses Wilson intervals.

10. Results and report-versus-code questions

Source: current calculations; primary_tests_adults.csv; sensitivity_tests_unadjusted.csv; REPORT_COMPARISON.md; report pages 7-10.

Factor	Adult comparison	Current raw p
Age	Stroke / no-stroke: 68.21 / 49.09 years	7.25356e-75
Glucose	Stroke / no-stroke: 133.10 / 107.00 recorded units	3.59113e-10
BMI	Stroke / no-stroke: 30.47 / 30.43 kg/m2	0.932776
Hypertension	With / without hypertension: 13.28% / 4.82%	3.43542e-14
Known smoking	Three-category chi-square test	0.00717472

65. What is the main age and glucose finding?

Answer: Adult stroke records have a mean age about 19.12 years higher and a mean glucose about 26.10 recorded units higher. Both raw tests are significant at 0.05. These comparisons are unadjusted for other characteristics.

66. How should you explain the BMI result?

Answer: The observed adult mean difference is only about 0.04 kg/m2 and raw p = 0.932776. The code does not detect a mean difference. Missingness, possible nonlinear patterns and other variables limit interpretation.

67. What does the hypertension percentage compare?

Answer: 66 of 497 adults with hypertension have a stroke label, compared with 181 of 3,757 without hypertension. These give 13.28% and 4.82%. The denominator is all adults within each hypertension category, not all stroke records.

68. Why is the smoking conclusion sensitive?

Answer: The known-status adult test has raw p = 0.00717472, while the complete-adult test has raw p = 0.0944168. It crosses the 0.05 threshold when records are restricted. Former smokers are also older, so the groups are not directly comparable as if other factors were equal.

69. Why do the slide and code p-values differ?

Answer: The earlier report uses Holm-adjusted p-values. This code reports raw p-values. For example, smoking is 0.0143 in the report versus 0.00717472 here; complete-adult smoking is 0.1888 versus 0.0944168. Counts, means and proportions still match.

70. Does this code calculate effect sizes, intervals or independent predictors?

Answer: It reports raw mean differences, but no Hedges g, Cramer V or confidence intervals. It fits no multivariable model, so it cannot identify independent predictors. A statistically significant result also does not establish causation or individual future risk.

11. Practical demonstration and troubleshooting

Source: main.py; read_dataset(); export_tables(); compare_report(); requirements.txt and README.md.

71. How do you run the project?

Answer: Open the folder containing main.py, stroke.csv and requirements.txt. Install dependencies with python -m pip install -r requirements.txt, then run python main.py. Use the same Python interpreter for installation and execution.

72. Which libraries are required?

Answer: NumPy, Pandas, SciPy, Matplotlib and Seaborn are the five third-party dependencies listed in requirements.txt. os is a standard-library module and does not need separate installation.

73. What would you demonstrate during the viva?

Answer: Show option 1 for dataset information, option 2 for cleaning and populations, option 5 for observed BMI, option 7 for smoking handling, option 8 for the five tests, and option 10 for complete outputs. Close each plot window, then choose 0 to exit.

74. What if stroke.csv is missing, empty or invalid?

Answer: read_dataset() checks file existence, empty data and required columns. Validation checks values and categories. main() catches preparation errors, prints the issue and returns. Restore the supplied CSV beside main.py and inspect the reported problem.

75. Why might output saving fail on Windows?

Answer: An output CSV open in Excel can prevent overwriting, or the folder may not be writable. Close the output file, check permissions and run again. A missing package usually means installing requirements in the Python interpreter selected in VS Code.

76. Does the reference CSV supply results to the analysis?

Answer: No. report_reference_values.csv contains labeled numbers from the earlier report, used only after current results are calculated for comparison. If it is absent, the analysis continues without that comparison. Statistics and graphs still come from stroke.csv.

12. Rapid revision and demonstration notes

Use this page immediately before your viva. Keep dataset proportions, test p-values and prediction claims separate.

A short project explanation

"My project analyzes associations between recorded stroke and five health factors using 5,110 records. The primary analysis uses 4,254 adults. I validate the data, use observed BMI and known smoking categories for the relevant tests, compare continuous means with Welch tests, and examine categorical associations with chi-square tests. The current code reports raw p-values and includes sensitivity checks. It supports an educational analysis, not causal conclusions or individual prediction."

If the examiner asks...	Say this
Why does smoking have no missing cells?	Unknown is a string category; it is not NaN.
Do all tests use 3,256 adults?	No. That is only the complete-case sensitivity population.
Did you remove 594 glucose outliers?	No. They were flagged and retained.
Does $p < 0.05$ mean a large effect?	No. A p-value does not measure effect size.
Is $p = 0.9328$ proof that BMI is irrelevant?	No. The adult mean test lacks evidence of a difference.
Is current smoking protective?	This analysis cannot support that claim.
Does the current code use Holm or Wilson?	No. Those belong to the earlier advanced report.
Can you identify independent risk factors?	No. The comparisons are unadjusted for other covariates.

Useful output files to open during the demonstration

outputs/FINDINGS.md: calculated conclusions.
outputs/tables/data_audit.csv: cleaning and population counts.
outputs/tables/primary_tests_adults.csv: hypotheses and raw p-values.
outputs/tables/sensitivity_tests_unadjusted.csv: population comparisons.
outputs/tables/report_comparison.csv: current versus report values.
outputs/charts/: 10 individual figures and the dashboard.

Source notes

Primary sources: current stroke.csv and the four Python modules, with outputs checked against their calculations. Supporting sources: README.md, REPORT_COMPARISON.md, requirements.txt, TESTING.md, Stroke_Capstone_Report.pdf and Problem_Definition_and_Dataset.pdf. This guide does not treat form or submission instructions inside the PDFs as tasks.

Project folder: Projects/Python project/Stroke_Analysis_Project/Stroke_Analysis_Project. Report folder: Projects/Python project/Documents. The supplied CSV has SHA-256 74a80e8ed80a03b7eb0976c3954ce3a5edd86d435cfff9b842d7a3a39b91ae93. This hash identifies file bytes, not clinical provenance.